18-09-2026

Business Intelligence

Executive Summary :-

⇒ This report delivers a rigorous risk and return analysis of five prospective funds under consideration for our next client offering. To eliminate subjectivity, all allocations are derived purely from the Sharpe Ratio.

⇒ The ultimate recommendation is a highly structured, risk optimized Three-fund Portfolio that achieves a Portfolio sharpe ratio of 2.97, significantly outperforming any individual fund on a risk-adjusted basis.

Expected Return : $E(R) = \frac{\Sigma(\text{Return})}{5}$

Standard deviation : $\sigma = \sqrt{\frac{\Sigma(R - E(R))^2}{5}}$

Greentech ventures :

$$E(R) = \frac{18 - 6 + 32 + 22 + 14}{5} = \frac{80}{5} = 16.00\ \%$$

$$\text{Variance} = \frac{(18-16)^2 + (-6-16)^2 + (32-16)^2 + (22-16)^2 + (14-16)^2}{5}$$

$$= \frac{784}{5} = 156.8$$

$\sigma = \sqrt{156.8} = 12.52\%$

$$\text{Sharpe} = \frac{16.00 - 4.00}{12.52} = 0.96$$

2. Steady state bonds:

$$E(R) = \frac{6+5+7+6+8}{5} = \frac{32}{5} = 6.40\%$$

$$\text{Variance} = \frac{(6-6.4)^2 + (5-6.4)^2 + (7-6.4)^2 + (6-6.4)^2 + (8-6.4)^2}{5}$$

$$= \frac{5.2}{5} = 1.04$$

$$\sigma = \sqrt{1.04} = 1.02\%$$

$$\text{Sharpe} = \frac{6.40 - 4.00}{1.02} = 2.35.$$

3. Metro Realty & Trust :-

$$E(R) = \frac{10+14-3+19+12}{5} = \frac{52}{5} = 10.40\%$$

$$\text{Variance} = \frac{(10-10.4)^2 + (14-10.4)^2 + (-3-10.4)^2 + (19-10.4)^2 + (12-10.4)^2}{5}$$

$$= \frac{269.2}{5} = 53.84 \Rightarrow \sigma = \sqrt{53.84} = 7.34\%$$

$$\text{Sharpe} = \frac{10.4 - 4.00}{7.34} = 0.87$$

4. Consumer Staples fund:

$$E(R) = \frac{9+11+8+10+9}{5} = 9.40\%$$

Variance = $\frac{(9-9.4)^2+(11-9.4)^2+(8-9.4)^2+(10-9.4)^2+(9-9.4)^2}{5}$

$= \frac{5.2}{5} = 1.04 \Rightarrow \sigma = 1.02\%$.

Sharpe $= \frac{9.40-4.00}{1.02} = 5.30$

5. Frontier Markets fund :-

$E(R) = \frac{25-15+40+5+30}{5} = 17.00\%$.

Variance $= \frac{(25-17)^2+(-15-17)^2+(40-17)^2+(5-17)^2+(30-17)^2}{5}$

$= \frac{1930}{5} = 386$

$\sigma = \sqrt{386} = 19.65\%$.

Sharpe $= \frac{17.00-4.00}{19.65} = 0.66$

PART 1: Individual fund analysis & Performance metrics.

Rank of Efficiency	Fund name	Expected Return (E(R))	Standalone Risk (σ)	Sharpe Ratio
Elite	Consumer Staples Fund	9.40%	1.02%	5.30
Excellent	Steady state Bonds	6.40%	1.02%	2.35
Moderate	GreenTech Ventures	16.00%	13.52%	0.96
Sub-par	Metro Realty Trust	10.40%	7.34%	0.87
Inefficient	Frontier Markets Fund	17.00%	19.65%	0.66

* Raw performance differs significantly from risk-adjusted performance. The frontier Markets fund has the highest expected return (17.00%) but the lowest sharpe ratio (0.66) due to its high volatility (σ = 19.65%). Consumer Staples has 9.40% returns with minimal volatility, making it the most attractive asset on a risk adjusted basis.

Part-2 Three fund Portfolio Optimization:

⇒ Selected Allocation :-

i. 70% Allocation (Maximum): Consumer Staples fund.

ii. 20% Allocation: Steady state bonds.

iii. 10% Allocation (Minimum): Metro Realty Trust

Portfolio Mathematical Expansion:

⇒ Using unit's required weighted-average methodology.

• Expected Return

$$E(R)_p = (0.70 \times 9.40\%) + (0.20 \times 6.40\%) + (0.10 \times 10.40\%)$$

$$E(R)_p = 6.58\% + 1.28\% + 1.04\%$$

$$= 8.90\%$$

• Combined Volatility:

$$\sigma_p = (0.7 \times 1.02\%) + (0.20 \times 1.02\%) + (0.1 \times 7.34\%)$$

$$\sigma_p = 1.65\%$$

• Final Sharpe Ratio:

$$\text{Sharpe} = \frac{E(R)_p - \text{Risk free rate}}{\sigma_p}$$

$$= \frac{8.90\% - 4.00\%}{1.65\%}$$

$$= \frac{4.90\%}{1.65\%} = 2.97$$

Part: 3

Investment committee Pitch:

"Mister chairman, members of the committee, I am presenting a mathematically optimized three fund portfolio consisting of 70% Consumer staples fund, 20% steady state bonds, 10% Metro Realty Trust. The firm's required framework, the asset mix generates a Expected return of 8.90% against a remarkably low volatile profile of 1.65%, producing an industry-leading sharpe Ratio of 2.97.

To maximize Risk-adjusted performance, we intentionally excluded GreenTech Ventures and Frontier Markets Fund. While frontier Markets display the highest standalone raw expected return at 17%.

its extreme 19.65% standard deviation creates an inefficient risk-per-unit-of-return tradeoff. Similarly, Greentech ventures introduced unacceptable volatility of drawbacks that drag down overall portfolio efficiency.

By placing 70% of our categorical capital into Consumer Staples, we are concisely accepting a concentration risk tied to a single defensive equity sector. While consumer staples protected capital perfectly for last 5 yrs, a systematic supply chain shock or sudden raw material inflation across consumer goods could expose this portfolio to downside vulnerability.

The 2.97 sharpe efficiency this calculated exposure provides our clients with the most mathematically sound product offering available